Advanced Algorithms 2026 19/10/2026

Exercise 05

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1 Warm-Up

Let $\phi^* = \min_{C \subseteq \mathbb{R}^d, |C|=k} \phi(X, C)$ be the cost of the optimal k-means solution, where centers C can be any k points in $\mathbb{R}^d$. Let $\hat{\phi}^* = \min_{C \subseteq X, |C|=k} \phi(X, C)$ be the cost of the optimal solution where the centers C are restricted to be a subset of the data points X.

Show that restricting the centers to X gives a 2-approximation, i.e.,

$$\hat{\phi}^* \leq 2\phi^*.$$

2 Analyzing the Final Step of k-means||

In this exercise, we analyze the final step of the k-means|| algorithm, where we reduce an oversampled candidate center set to exactly k centers using weighted k-means++.

Throughout, $X \subseteq \mathbb{R}^d$ is a dataset, C^* denotes an optimal set of k centers, and $\phi^* := \phi(X, C^*)$ denotes the optimal k-means cost.

Let $B \subseteq X$ be the (possibly large) set of candidate centers produced by the k-means|| overseeding procedure. For each $x \in X$, let $\pi(x) \in B$ be a closest point in B to x, breaking ties arbitrarily. For each $b \in B$, define the *weight*

$$w_b := \big|\{x \in X : \pi(x) = b\}\big|.$$

Let $\phi_w(B, C)$ denote the weighted k-means cost:

$$\phi_w(B, C) := \sum_{b \in B} w_b \cdot \min_{c \in C} \|b - c\|^2.$$

Exercise 2A

Show that for every $x \in X$,

$$\min_{c \in C} \|x - c\|^2 \;\leq\; 2\|x - \pi(x)\|^2 \;+\; 2 \min_{c \in C} \|\pi(x) - c\|^2.$$

Hint. Use the squared triangle inequality:

$$\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2.$$

Exercise 2B

Show that for any $C \subseteq \mathbb{R}^d$,

$$\phi(X, C) \;\leq\; 2\phi(X, B) \;+\; 2\phi_w(B, C).$$

Exercises 2C-2E

For the following exercises, suppose we are given the two results from the k-means|| procedure:

(A1) The expected overseeding cost is bounded by the optimal k-means cost:

$$\mathbb{E}[\phi(X, B)] \leq \alpha \cdot \phi^*, \quad \text{for some } \alpha \geq 1.$$

(A2) The weighted k-means++ step returns a set C satisfying a β-approximation guarantee in expectation for the weighted problem (B, w):

$$\mathbb{E}[\phi_w(B, C) \mid B] \leq \beta \cdot \min_{\substack{C' \subseteq \mathbb{R}^d \\ |C'|=k}} \phi_w(B, C'), \quad \text{for some } \beta = O(\log k).$$

Exercise 2C

Show that:

$$\mathbb{E}[\phi(X, C) \mid B] \;\leq\; 2\phi(X, B) \;+\; 2\beta \cdot \phi_w(B, C^*).$$

Exercise 2D

Show that:

$$\phi_w(B, C^*) \;\leq\; 2\phi(X, B) \;+\; 2\phi^*.$$

Hint. Use the definition of the weights w_b to rewrite $\phi_w(B, C^*)$ as $\sum_{x \in X} \min_{c \in C^*} \|\pi(x) - c\|^2$. Then apply the squared triangle inequality with $u = \pi(x) - x$ and $v = x - c$ for $c \in C^*$.

Exercise 2E

Show that

$$\mathbb{E}[\phi(X, C)] \;\leq\; f(\alpha, \beta)\, \phi^*$$

and determine the function $f(\alpha, \beta)$.

3 Settled Clusters and k-means||

This exercise asks you to prove a key lemma in the analysis of the k-means|| overseeding algorithm.

Definitions. Fix an optimal solution $C^* = \{c_1^*, \ldots, c_k^*\}$ with clusters $A_1^*, \ldots, A_k^*$ and optimal cost $\phi^* = \sum_{i=1}^k \phi(A_i^*, \mu_{A_i^*})$.

For a current center set C, we say an optimal cluster A_i^* is:

- *settled* if $\phi(A_i^*, C) \leq 10\, \phi(A_i^*, \mu_{A_i^*})$
- *unsettled* otherwise.

In one round of k-means||, each point $x \in X$ is added to the center set *independently* with probability $p(x) = \min\left\{1,\ \ell \cdot \frac{\phi(x,C)}{\phi(X,C)}\right\}$. We say A^* is "made settled" if, after this round, it is no longer unsettled.

Fact (Given Lemma). You may use the following lemma from the lecture notes without proof: Let $c \in A^*$ be a random variable drawn from the distribution $\Pr[c = x] = \frac{\phi(x,C)}{\phi(A^*,C)}$ for $x \in A^*$. Then,

$$\mathbb{E}\big[\phi(A^*, C \cup \{c\})\big] \leq 8\, \phi(A^*, \mu_{A^*}).$$

Exercise 3A

Let $A' \subseteq A^*$ be the set of "good" points that are sufficient to settle the cluster A^*:

$$A' := \{x \in A^* \mid \phi(A^*, C \cup \{x\}) \leq 10\,\phi(A^*, \mu_{A^*})\}\,.$$

Show that

$$\phi(A', C) \;\geq\; \tfrac{1}{5}\,\phi(A^*, C).$$

Exercise 3B

Let C be the current center set and A^* be an unsettled cluster. Show that the probability that A^* is *not* made settled in the next round is at most

$$\exp\!\left(-\frac{\ell\,\phi(A^*, C)}{5\,\phi(X, C)}\right).$$